

BAF53b Deletion in Parvalbumin Interneurons Impacts the Temporal Dynamics of Social Communication in Neonatal Mice

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Background & Motivation

Purpose: Autism Spectrum Disorder (ASD) is one of the most common neurodevelopmental disorders, with differences in social communication being a major characteristic of individuals with ASD. The specific mechanisms behind these differences remains unknown.

Ultrasonic Vocalizations (USV)

- Proxy for social communication
- Signals distress or isolation to mothers

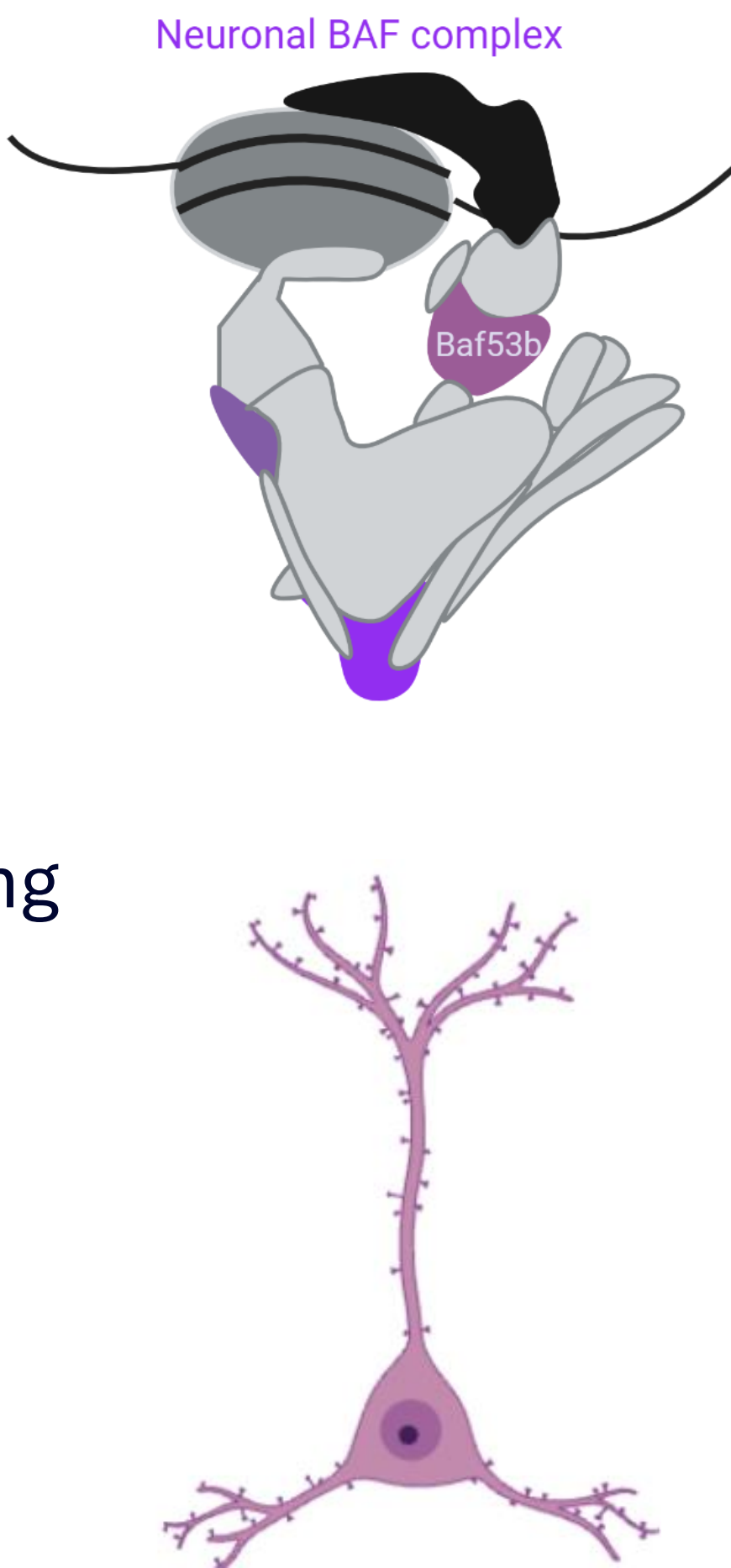
BAF53b

- Subunit of the nBAF chromatin-remodeling complex
- Regulates chromatin accessibility, impacting downstream gene expression

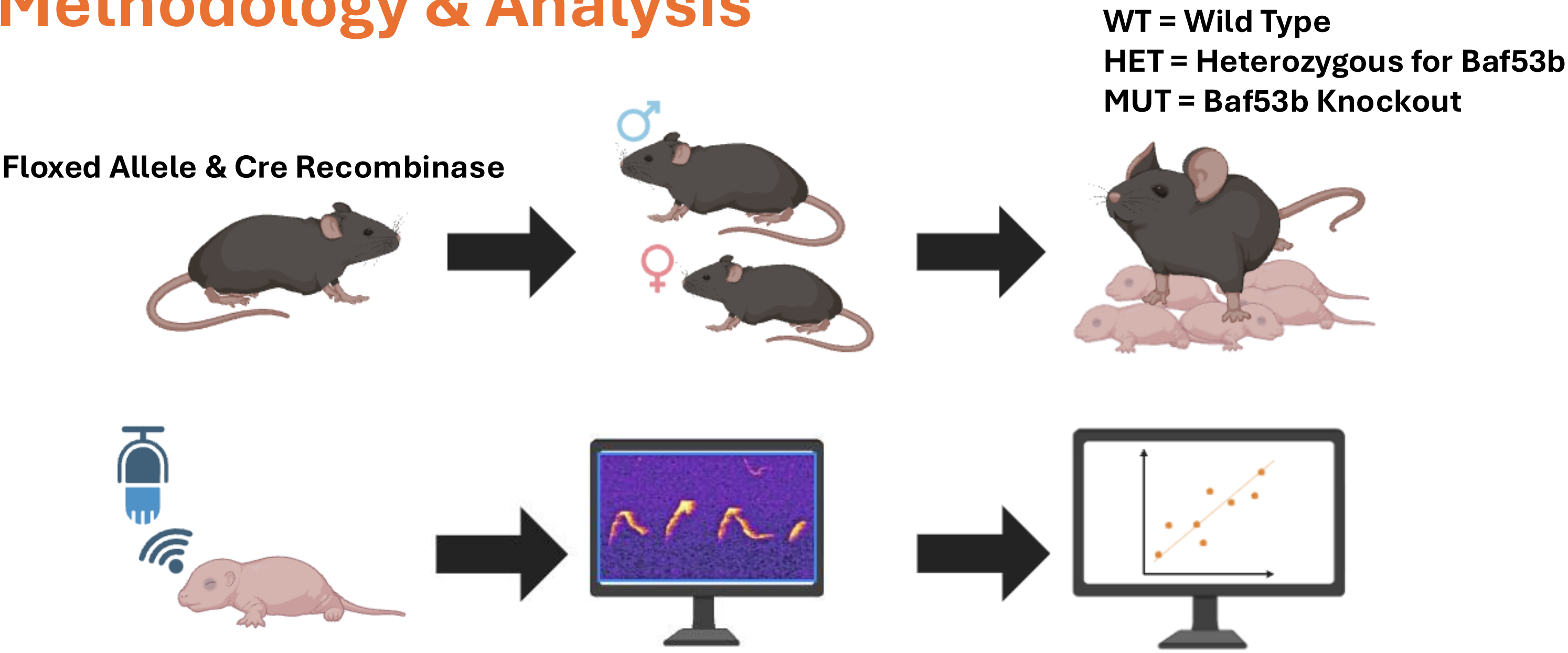
Parvalbumin Interneurons (PV)

- Subtype of GABAergic inhibitory neurons
- Support E/I circuitry balance and neuronal timing
- Altered inhibitory circuitry is a recurring theme in ASD models

Hypothesis: BAF53b deletion in mice PV interneurons impacts E/I circuit maturation, causing **changes to USV regulation** in BAF53b mutant and heterozygous mice models.

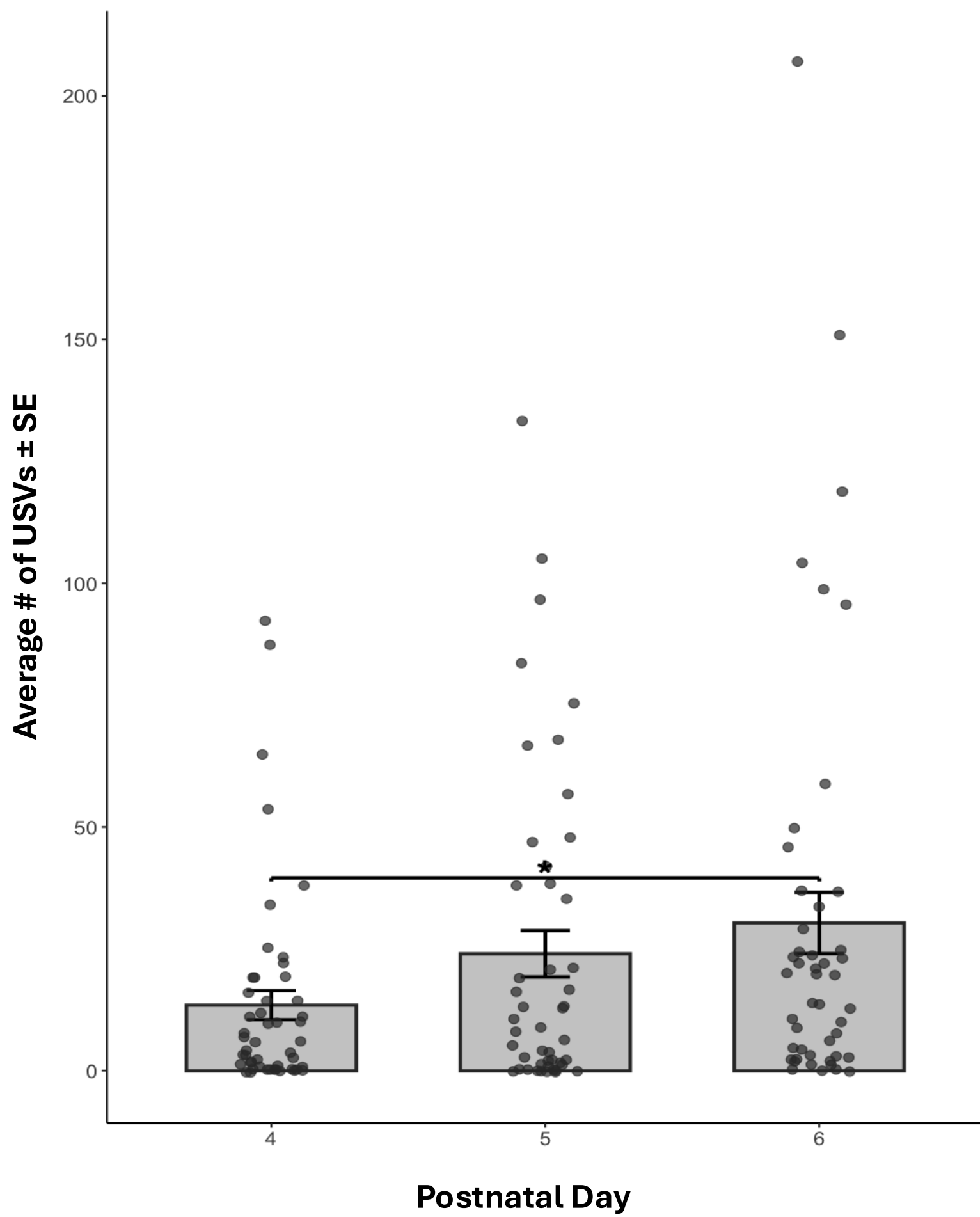


Methodology & Analysis

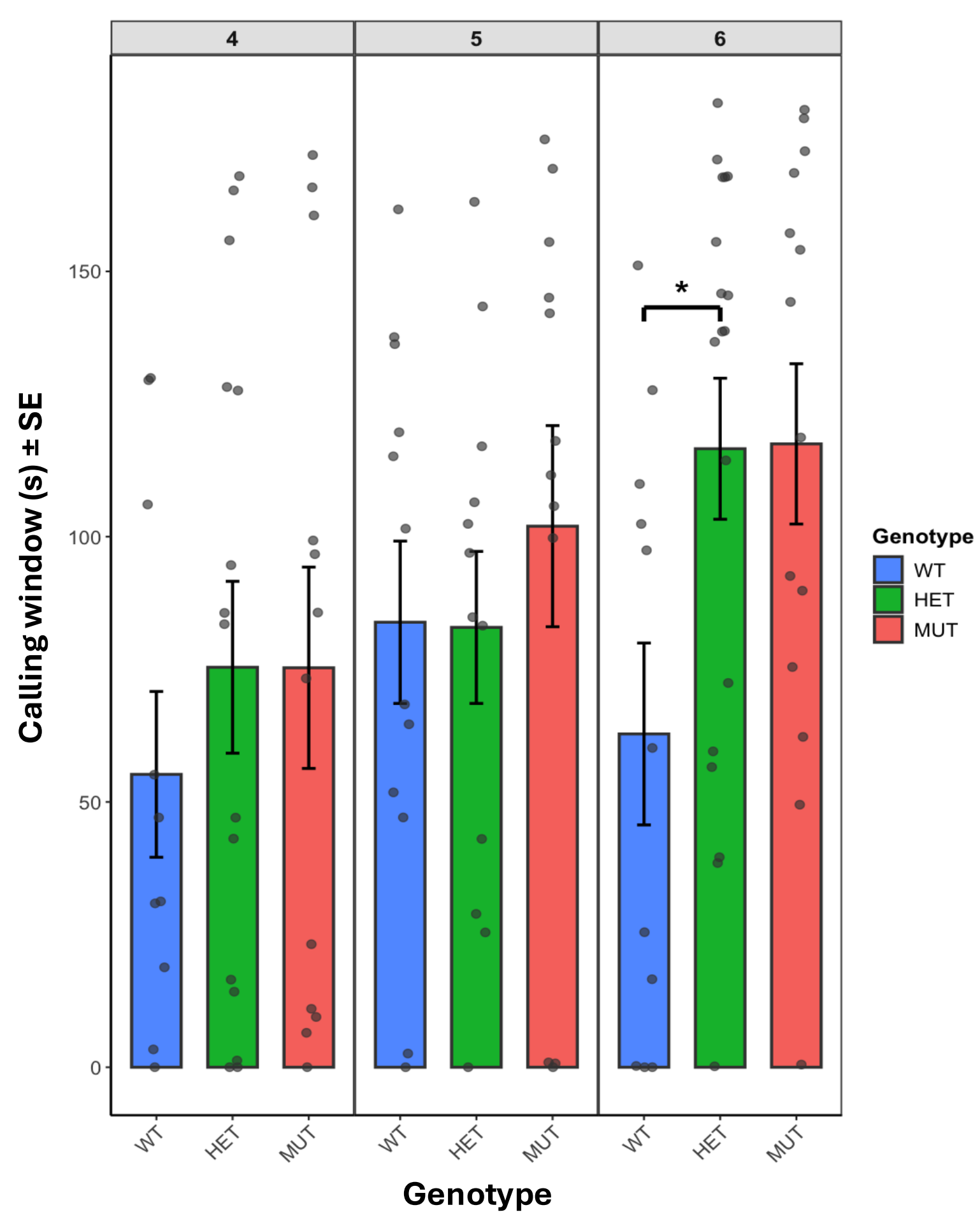


Results & Discussion

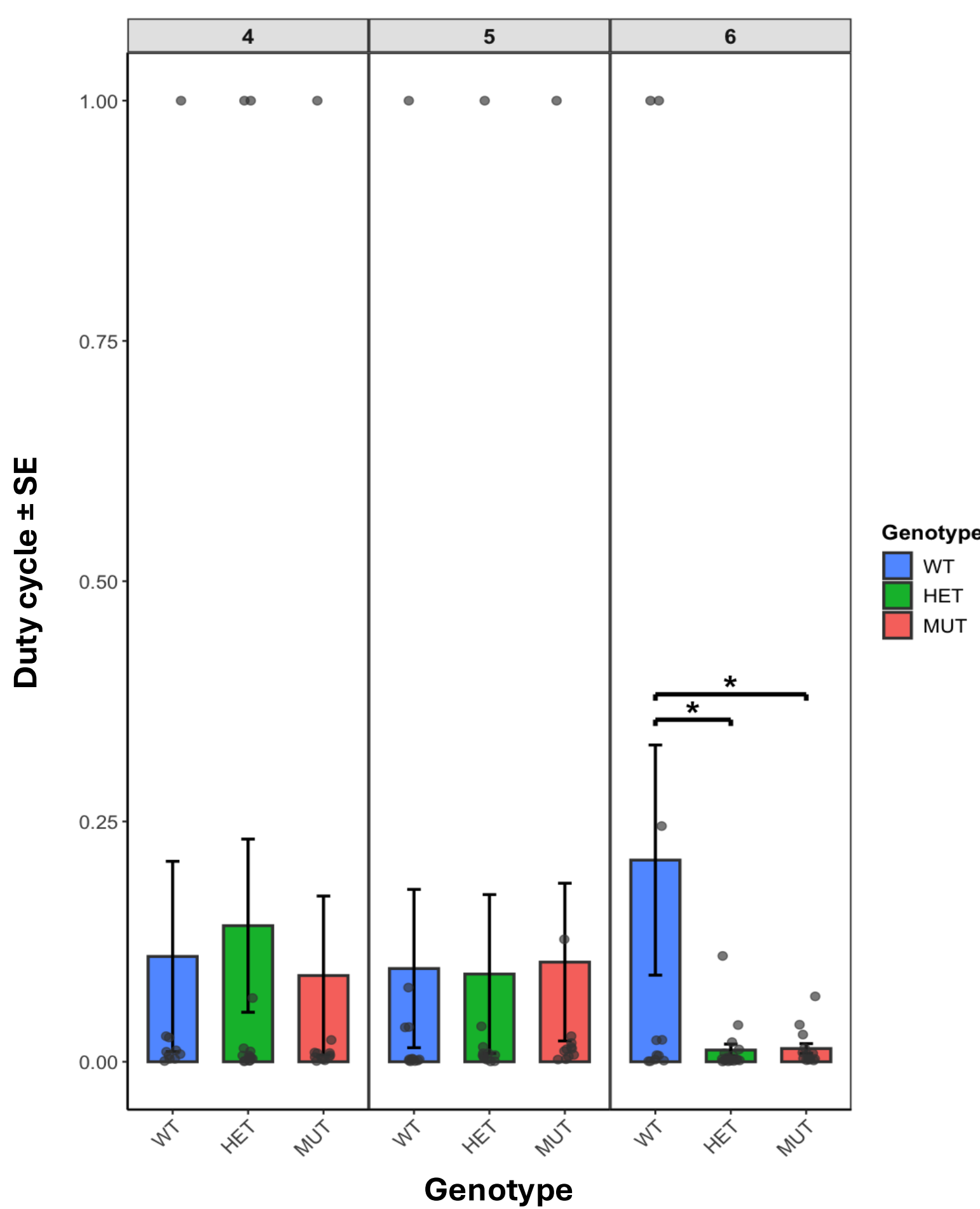
Average number of USVs across day



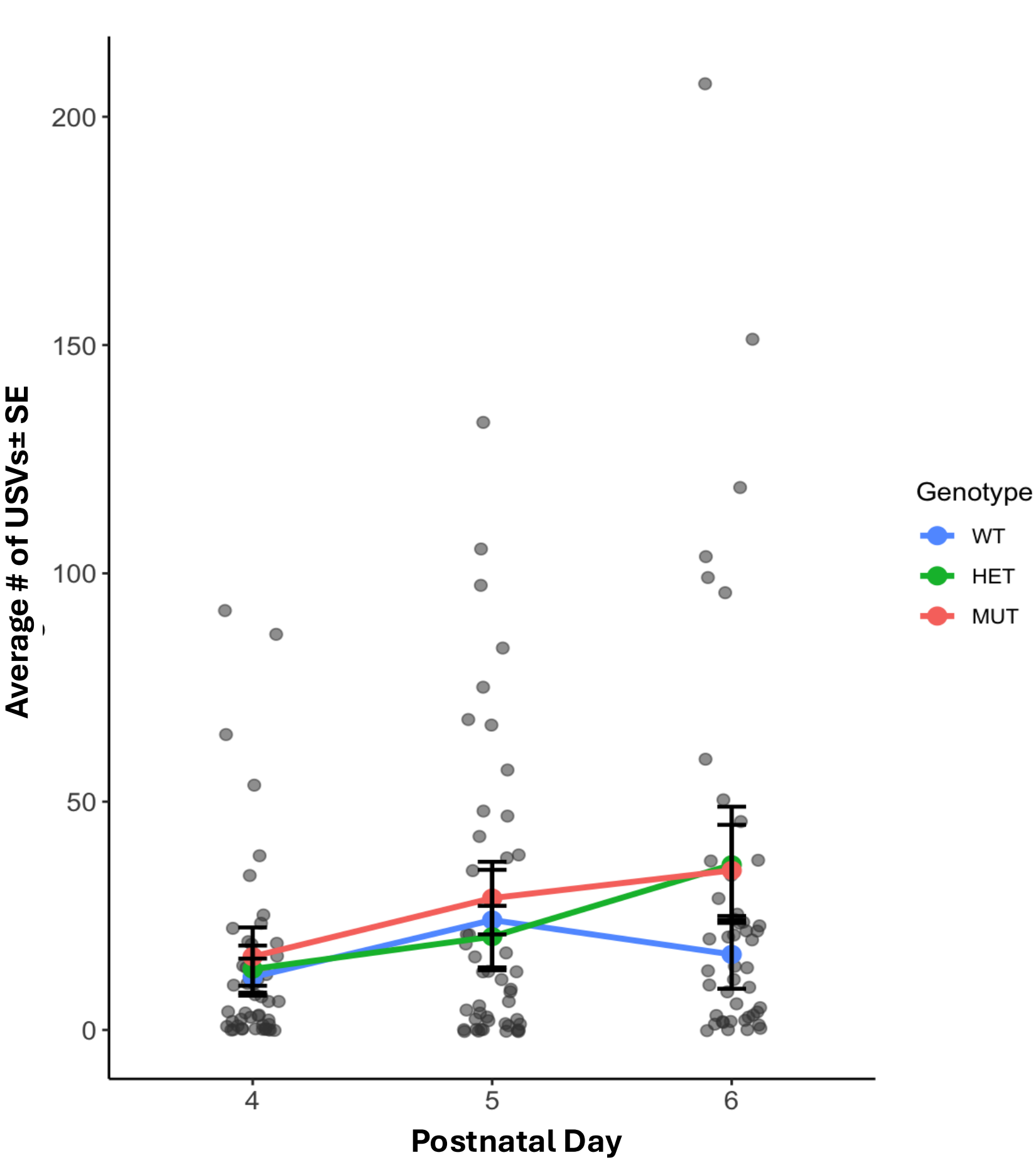
Calling Window by genotype across day



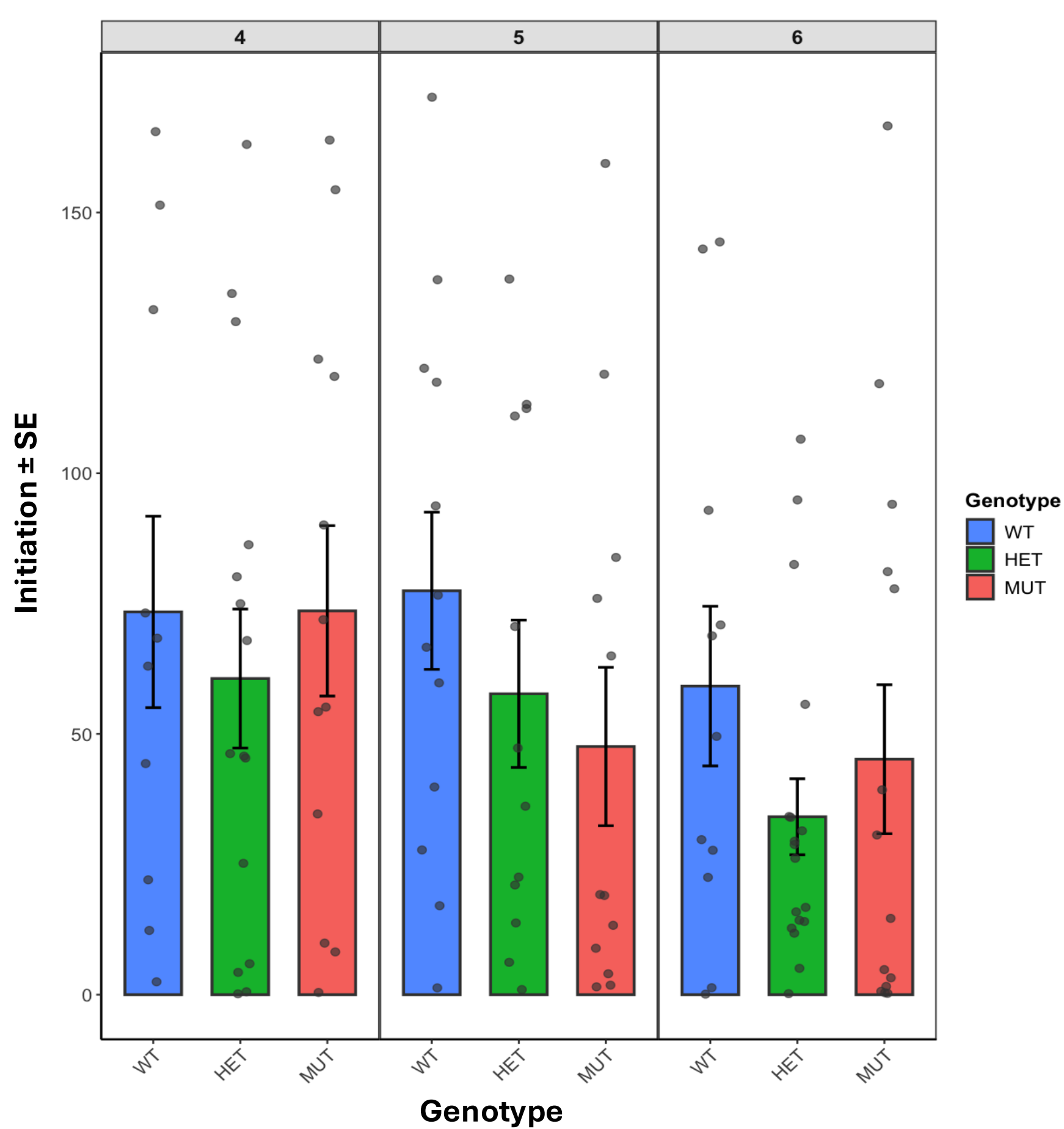
Proportion of Call Time in Calling Window by genotype across day



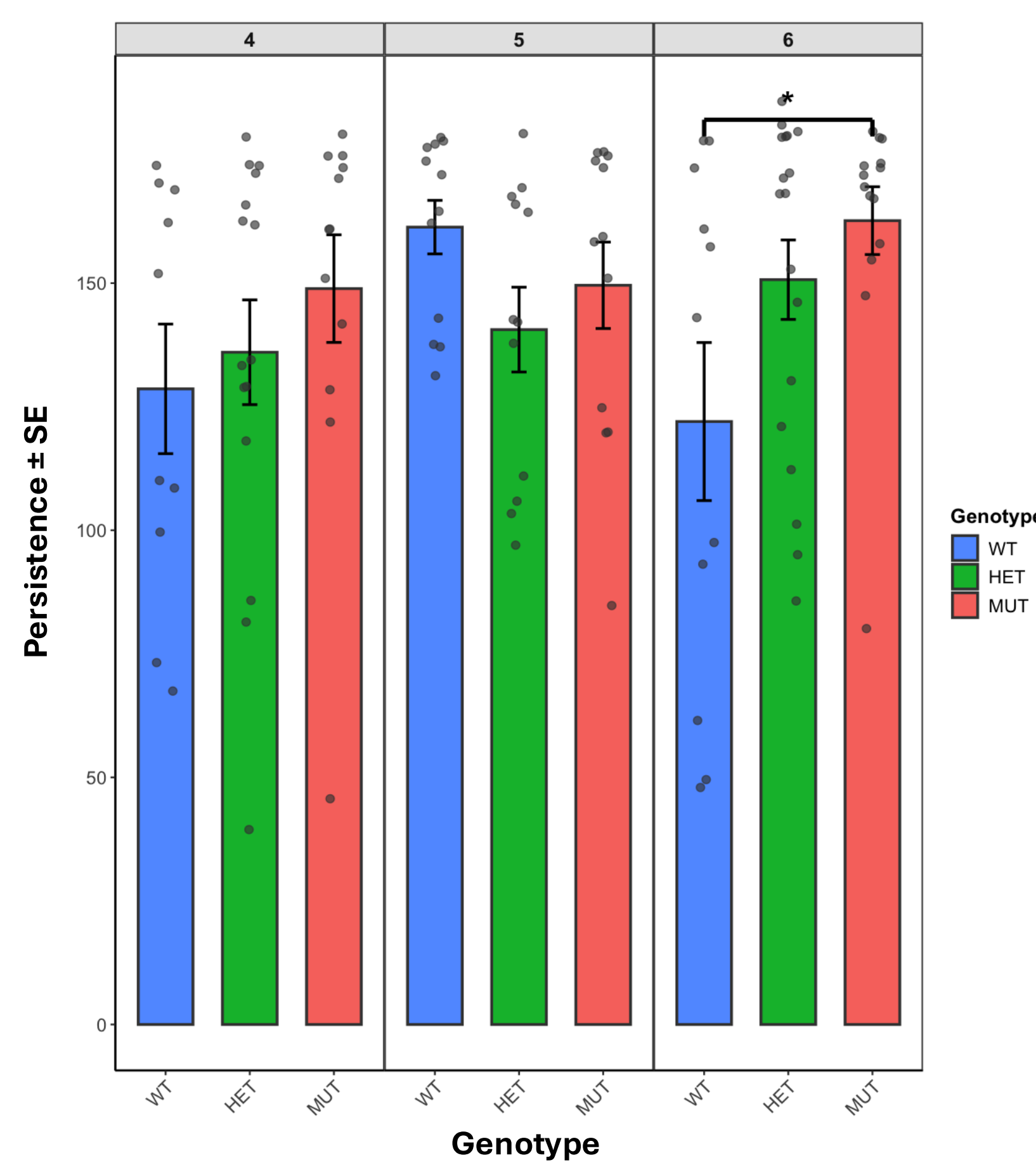
Average number of USVs by genotype across day



Latency to First Call by genotype across day



End Time of Last Call by genotype across day



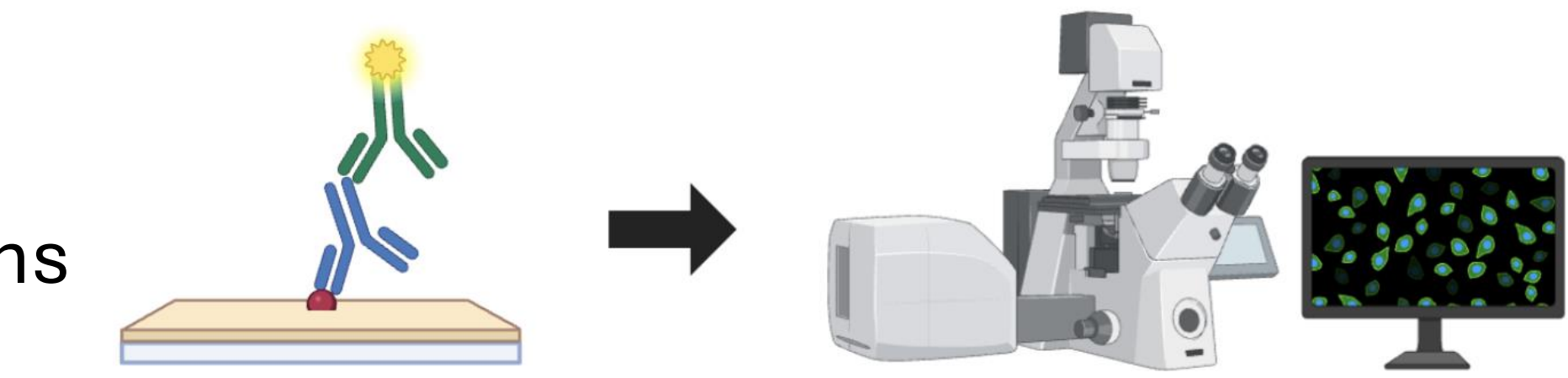
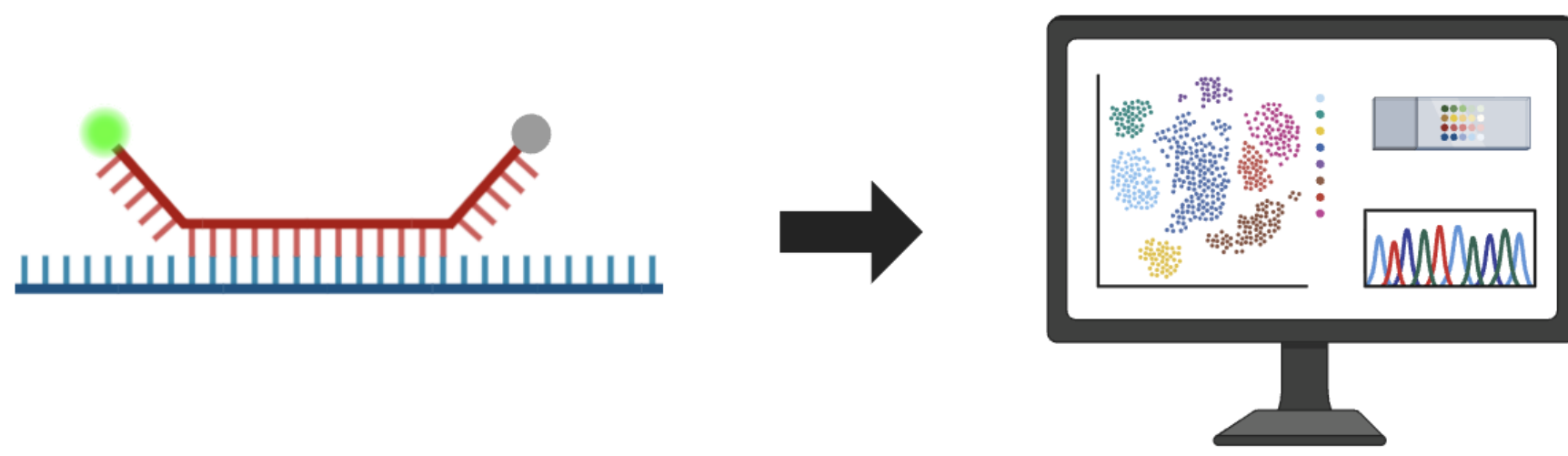
Conclusion

BAF53b deletion increases the call window length through a later end time of last call, and decreases the proportion of time spent calling in the window, both at Postnatal Day 6.

Future Directions

Immunostaining & Confocal Microscopy

- Changes in Presynaptic and Postsynaptic Proteins
- Alterations in Inhibitory Synapse Circuitry



RNA scope & Spatial Transcriptomics

- Quantification of RNA molecules
- Spatial Gene Expression